

Update Report

Progress Report and Project Status Update for CSE445

1. Executive Summary

This update report documents the progress of the CSE445 project "Stop Sign Detection with YOLOv8" between the interim submission and the final submission. The project reached its final stage: the model has been trained from scratch on a custom dataset, evaluated, and deployed as an easy-to-use application. All deliverables described in this update - the final report, the final presentation, the update presentation, and the demo video - are ready for the final submission.

2. Work Completed Since the Last Update

2.1 Dataset Finalization

The dataset was finalized at 65 street photographs containing stop signs. Every image was annotated with a tight rectangle using labelme and converted to the YOLO text format. The dataset was split into 55 training images and 10 validation images. The split was programmatically audited to guarantee that no image appears in both subsets, eliminating any possibility of data leakage in the reported metrics.

2.2 Model Training and Improvement

An initial model trained on a 50-image dataset reached a mAP@0.5 of only 0.856. After growing the dataset to 65 images, correcting annotation issues, and establishing a clean train/validation split, the final from-scratch YOLOv8n model reached a mAP@0.5 of 0.995 with a recall of 1.000. Training was completed on CPU in about 61 minutes (214 epochs with early stopping).

2.3 Application and Sharing

A user-friendly application was completed: python main.py processes the validation set or arbitrary photos, and test_photo.bat provides a graphical double-click tester with a file picker. The project was packaged with setup.bat and requirements.txt so a friend can install and run it with standard Python.

3. Key Results

The final model achieves a precision of 0.957, a recall of 1.000, an F1 score of 0.978, a mAP@0.5 of 0.995, and a mAP@0.5:0.95 of 0.712 on the 10 held-out validation images. Every validation image was detected correctly with exactly one tight bounding box, a per-image accuracy of 100%.

4. Challenges and How They Were Solved

Several challenges were overcome during the project: the automated image downloader was blocked by the search engine, so images were collected manually; the first annotation tool crashed, so labelme was used instead; CPU-only training was slow, so the smallest model variant and early stopping were used; and an inference-resolution mismatch caused missed detections until the input size was aligned to the training size. Each issue was diagnosed and

resolved, and the solutions are documented in the final report.

5. Deliverables for Final Submission

Final report (8-page IEEE-format PDF), final presentation (PPTX), update report (this document), update presentation (PPTX), one-minute demo video, the full source code, the dataset, and the trained model. All files are organized in the GitHub repository according to the course guidelines.

6. Next Steps

The remaining steps are to present the project during the final presentation session and to submit the repository with all files before the deadline.